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Industrial Data Reduction, Aggregation and Machine Learning-Based Soft Sensing for Etching and Slider Production Tools

A thesis submitted in partial satisfaction
of the requirements for the degree Master of
Science in Chemical Engineering

by

Julius Owen Suherman

2025

ABSTRACT OF THE THESIS

Industrial Data Reduction, Aggregation and Machine Learning-Based Soft Sensing for Etching and Slider Production Tools

by

Julius Owen Suherman
Master of Science in Chemical Engineering
University of California, Los Angeles, 2025
Professor Panagiotis D. Christofides, Chair

SmartManufacturing(SM),whichisshortfor“Smart(Predictive,Preventive,Proactive) zeroincident,zeroemissionsManufacturing,”describesmanufacturing’sdigitaltransformation in which factories, supply chains and ecosystems are integrated, interoperable, and interconnected. Smart Manufacturing is rooted in AI, Machine Learned (ML), and Data Synchronized (DS) model-ing to tap into invaluable operating data. By making data actionable at larger scales, SM opens new ways to increase productivity,precision,and process performance.Smart Manufacturing applied tofront-endwafermanufacturinginthesemiconductorindustryofferssignificantopportunityto increase production throughput and ensure precision by increasing staff and operational productiv-ity.Front-end wafer manufacturing involves multi tool operations for complex material processing thatrequiresahighdegreeofprecisionandextensiveproductqualification.Thereisahighde-gree of commonality with semiconductor manufacturing tools, for example etching, that are well instrumented.Companies are already collecting large amounts of operational data from these tools

that can be aggregated and leveraged for virtual metrology and other control, diagnostic, and man-agement solutions.AI/ML/DS modeling involves monitoring the state of an operation in real-time to continuously learn and improve on human centered, automated, and autonomous actions.This operational data are embedded in invaluable machine, process, product, and material behaviors as interaction complexities,linearities/non-linearities,and dimensional effects.Because of machine commonalities, data can be selected to draw out operational value across machines.Today's data science offers considerable capability for qualifying, assessing alignment and contribution, aggre-gating, and engineering data for more robust modeling.We refer to this as a Data-first strategy to process,engineerandmodelwithAI-Readydata.Inthispaper,weaddressAI-Readydatafora virtual metrology solution focused on etching measurement PASS/FAIL classification and milling depthpredictionregressiontasksusingoperationaldatafromproductionmachinetools.Ifthe quality of the product can be predicted, the productivity of the metrology process can be increased, whichinturnincreasestheproductivityoftheoveralloperation.Inapreviouspaper,wecon-sidered how to aggregate data from different etch tools in the same processes at different factories within Seagate Technology and proposed a method for data aggregation and demonstrated its value [1].The present paper considers how to process and engineer datasets from two different etch tool processes:wafer and slider production.The data processing approaches when used systematically with appropriate ML algorithms demonstrate the potential for reducing metrological interventions insemiconductormanufacturing.Advancedmachinelearningtechniquesareusedtotacklethe modelingchallengesofalowfailurerateandlimitedoperationalvariability.XGBoost,agradi-ent descent-based tree algorithm, outperforms the commonly used Feedforward Neural Networks (FNN) in terms of training speed and resource utilization for binary-classifications. Principal Com-ponentAnalysis(PCA)effectivelyreducesthedimensionalityofthedataandoverfitting,while retainingvitalvariancesandsignificantlyreducingnoise.Dataaggregationwithseparatedscal-ingharmonizesinputsfromdiversemanufacturingtoolsandsignificantlyimprovestheefficacy andversatilityofcombiningmultipledatasetstoimprovemodelperformance.Aliveupdating

transfer learning approach, that periodically updates the FNN models in real-time using Stochastic GradientDescent(SGD)withindividualdatapoints,addressesprocessdrift,andmarkedlyim-proves predictive accuracy.For the slider production tools, data augmentation with linear Mixup, overcomesashortrecordingperiod,enrichesthetrainingdataset,andsignificantlyreduceserror metrics.

The thesis of Julius Owen Suherman is approved.

Dante A. Simonetti

Philippe Sautet

Panagiotis D. Christofides, Committee Chair

University of California, Los Angeles

2025

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Chapter 1

Introduction

As the transition to the digital era accelerates and the Internet of Things (IoT) grows, the demand for microelectronic devices is skyrocketing.The Semiconductor Industry Association (SIA) announced that global semiconductor industry sales totaled $574.1 billion in 2022, the highest ever annualtotalandanincreaseof3.3%comparedtothe2021totalof$555.9billion.Theindustry shipped a record 1.15 trillion semiconductor units in 2021, as chip companies ramped up production to address high demand amid the global chip shortage [2].These are devices equipped with integrated circuits such as central processing units (CPUs), graphics processing units (GPUs), hard drives,andsolidstatedrives[3].Demandforthesedeviceshaslongdriventheneedforhigher transistor densities and narrower gate widths to improve computing performance and reduce power consumption rates [4].Today, the increase in demand has led to recurring shortages [5], negatively affectingglobaleconomiesinvarioussectors,forexample,theautomotiveindustry[6].There is a substantial incentive in semiconductor manufacturing to increase production volume, increase product quality and precision, and increase productivity without only increasing operational equip-ment and personnel.Digital transformation and leveraging Industry 4.0 for Smart Manufacturing arepivotalwithhowtotakeadvantageofoperationaldata,AI/ML,IoT,informationtechnology, andinterconnectednessatgreaterscalestoopennewavenuesofincreasedproductivity,perfor-

manceandprecision[7].Smartmanufacturingis,bydefinition,aboutusingreal-timedataand modelingatscale.Itistheintersectionofplant-wideagilityandoptimization,sustainablepro-duction and resilient demand-driven supply chains made interoperable through interconnectedness with trust and meaningfully shared data and tools [8].

Smart Manufacturing at scale depends on operating sensor data collected, contextualized, and aggregated from sensors on factory floor operations. Specifically, sensors are installed on each ma-chine for each process operation, and they process and/or collect data in real time.These data are used to manage,control,and optimize the performance of each individual machine tool and pro-cess operation.They are useful for multiple management objectives such as monitoring, diagnosis, operationalhealth,preventivemaintenance,fasterchangeoversandqualityassurance[9].These dataarecriticaltoautomationandautonomousoperations.Whenusedacrosslineandfactory operations, the data are used to address interoperability, agility, and higher-level key performance indicatorsincombination.Lineoperationandfactoryinteroperabilityextendintosupplychain interoperability.Cross company data that provide visibility into supply chain material and product flowsandproductdemandsandcapacitiesareneededforresilience.Inadditiontothebenefits of applying data and models to improve physical operations at all levels, these same data are also valuable assets that can be aggregated to build richer data sets for model building [1].The data can be categorized,discovered and used to validate models and when synchronized with models,the resultingdigitaltwinsystemscanbeusedforreal-timepreventive,proactive,predictivecontrol, management and optimization [10].

Inthispaperwefocusondataasvaluableassetsanddemonstratehowdatafrommultiple similarmachinescanbeaggregatedintoricherdatasetsformorerobustmachinelearning(ML) modelbuildingforallmachinesand/orhowmultiplemachinescanbeoptimizedtogether.We focusonmethodanddemonstrationofvirtualmetrology(a.k.a.,softsensing)asoneimportant application in semiconductor manufacturing.Unlike direct measurement methods, such as an of-flinequartzmicrobalancetoanalyzelayerthicknessmanufacturing[11],MLvirtualmetrology

predicts product quality measurements or condition from operating sensor data by using these data totrainmodelsthatcan,withintherangeofoperatingconditionsexperienced,associateoperat-ingconditionstoqualitymeasurement[12].Theadvantageofsoftsensingliesinitsabilityto overcomethedrawbacksofdirectsensing,whichcanbeexpensive,time-consuming,untimely and/or labor-intensive [13].There can be situations in which the measurement costs more than the product.There are also situations in which technical measurement complexity, product conditions, and or measurement objectives make physically impossible to do a direct measurement, e.g., key performance indicators (KPIs).

In contrast, virtual metrology uses data generated during production processes. These are data thatembedoperationaleffectsfortherangeofoperationsexperienced.Thedata,collectedfrom widely available sensors are relatively inexpensive.**The engineeirng of the data and the modeling** process, if done systematically and carefully, has been shown to be highly effective in measuring physical parameters that are difficult to directly assess [14].Benefits from these models can accrue quickly.However,theseMLmodelsarelimitedtooperatingrangeexperienced.Itistherefore crucial to pay close attention to the operating conditions within which the reliability of the model ishigh.Modernmanufacturingprocesseshavebecomeincreasinglycomplex,generatingvast amountsofdatafromdozensorevenhundredsofsensors[15].Therecanbedatavolumesand operationalcomplexitiesthatexceedhumanassimilation.Machinelearninganddeeplearning techniqueshaveemergedaspromisingmethods.Theyperformwellincapturingcomplexnon-linear correlations between inputs and outputs, and it has been demonstrated that neural networks of sufficient scale can approximate any non-linear functions [16].

Severaltypesofneuralnetworkshavebeensuccessfullyimplementedinproductionset-tings:for example, Convolutional Neural Networks (CNNs) [17] and Recurrent Neural Networks (RNNs) [18].Transformer networks have been used by companies including Seagate Technology for fault detection in etch tools [19].Although recent advancements in ML have prompted exten-sive research into the application of soft sensing across various industries, challenges such as data

insufficiency,sensornoise,andthepresenceofredundantsensorsremainsignificantproblems. Furthermore,theissueofprocessdriftovertimecomplicatestheeffectivenessandreliabilityof models built on historical data.

Inthispaper,MLvirtualmetrologyistailoredforbothplasmaetchingandsliderproduction.Specifically,datasetsfrommultipleplasmaetchingandsliderproductiontoolswereconsistentlycollectedandcontextualizedbyapplyingCESMII'sDataInformationModelstructure, calledaProfile[8].TheProfileensuresthatthemeasurementsfromeachofthemachinescould beconcatenatedforfurthercross-machineanalysesandusesofthedata.Importantly,thepaper recognizesthattheseAI/ML/DSmodelsdepend on,infact thriveon the"right" data whenthere is enough that is engineered appropriately to build, train, test, and validate the AI/ML models.A"Data-First" strategy emphasizes the importance of data available in the amounts, forms, scale, and access necessary to achieve benefits in productivity, jobs, market share, sustainability, and growth. AData-FirststrategyemphasizesengineeringAI-Readydatathatisconsistently,collected,in-gested, contextualized, cleaned, normalized, protected, aggregated, assessed, and selected to draw out the operational value that has been refined from years of experience even when the physics may not be fully understood.Data-First also addresses the primacy of data access in implementing af-fordable AI/ML solutions including how to obtain, categorize, scale, and use AI-Ready data.This includes how to validate, learn, unlearn, and adjust models when using and reusing this invaluable data.

This paper explores the impact of data modification and manipulation techniques,including dimension reduction, data aggregation, and data augmentation, on model performance to address thesechallenges.PrincipalComponentAnalysis(PCA)isemployedasadimensionreduction technique to eliminate redundant features and reduce noise.Building upon a prior work [1],this study introduces a separate scaling method designed to normalize input datasets more effectively, thereby enhancing model performance.Despite the common use of Feedforward Neural Networks (FNN)insuchapplications,decisiontreemethodsbasedongradientdescentboosting,which

are well-suited for handling large feature sets with lower computational training costs, have rarely been explored in this context. To address the issue of process drift, a live-updating transfer learning approach is implemented with FNN models.This approach periodically updates the model using up-to-datedatatoreducetheneedforphysicalmeasurements,leveragingabasemodeltrained onhistoricaldata.Additionally,thispaperdiscussesandappliesadataaugmentationmethod, linear Mixup, to mitigate commonly encountered data insufficiency issues, which may arise from sensorfailuresorpre-matureprocesslines.TheMixupmethodhasbeenshowntosignificantly improve model performance as data-adaptive regularization, offering a promising solution to these pervasive challenges [20].

This work is organized as follows: Chapter 2 provides an overview of the datasets with Chap-ter 2.1 going more in depth on the preprocessing procedures for both etching and slider production tools, Chapter 2.2 presents the machine learning model construction for both classification and re-gression tasks, Chapter 3.1 and Chapter 3.2 demonstrate and discuss the model performance results on both tools, and Chapter 4 summarizes the findings of this work.

Chapter 2

Data Preprocessing and Modeling

This section provides an overview of the solution objective, data collection, physical functionality, and specific machine tools for the use case used in this study.As mentioned, the use case encom-passes five plasma etching machines at various Seagate factories used in wafer production distinct fromthoseexaminedinpreviousresearch[1],andanadditionalsevenslidertoolsfromthepro-duction of the slider component used in hard disk drives (HDD). The 12 datasets,one from each machine,made it possible to extend the prior study on data aggregation on similar machines and processes to a study on similar machines but different processes.The solution objective remained the same as the previous study [1], in which the model was constructed to use machine tool data to determine (predict, before metrology) if the final thickness of the deposited layer PASS or FAIL. PASS refers to if the wafer is expected with high probability to fall within a specified quality range, and FAIL is if it does not.As in the previous study, a classification model was applied.In addition, aregressionmodelwasalsodevelopedforthesliderproductionprocesstoquantitativelypredict thedepthofthemillingprocess.Bothmodelsaimtodeterminewhethertheoutputproductsof these tools are expected to meet specific metrological standards.If the performance of these mod-elsissufficientlyhigh,thisvirtualmetrologyapplicationcanincreasethemachineandstaffing productivity of the physical metrology quality assurance process.

The datasets produced by the etching process facilitate the development of a binary classification model to perform fault detection.This model classifies the final thickness of the deposited layer as PASS if it falls within a predefined range and FAIL if it does not.For the slider production process,a regression model is developed to predict the depth of the milling process.This model aimstodeterminewhethertheoutputsofthesetoolsmeetspecificmetrologicalstandards,thus servingasaneffectivesoftsensorforqualityassuranceformultipleproductsinmanufacturing processes.

Thedetailsofthedatacollectionanddescriptionofthetoolsandmodulesofdatasetswere elaboratedinapriorwork[1]anddemonstratedinFigure2.1.Insummary,themanufacturing processrequiresrawwaferstoundergoasequenceofprocessstepstotransformintocomplete products.Due to the repetitiveness of these processes, a wafer may interact with the same kind of tool multiple times at different processes.This work analyzes etching and slider production tools, whereeachtoolfunctionsasareactorandcontainsmodulesthatoperateasnearlyindependent reaction chambers to conduct specific processes.Tool-module combinations are designated in the T-PM format; for instance, ‘T1-PM1’ refers to module 1 in tool 1.For the rest of the paper, a data point is defined as a data vector of all features after averaging across the duration of the batch.A dataset is the collection of data points obtained from a specific tool-module combination.

2.1 Data Generation and Preprocessing for Etching and Slider Tools

2.1.1 Wafer Production Etch Tools

Operationaldatafromeachwaferproductionetchtoolwascollectedandconsistentlyorganized by the CESMII Information Model or Profile pictured in Figure 2.1.The data from each machine comprisedtwodatatypes:32streamed,time-basednumericalsensormeasurementsand2cate-

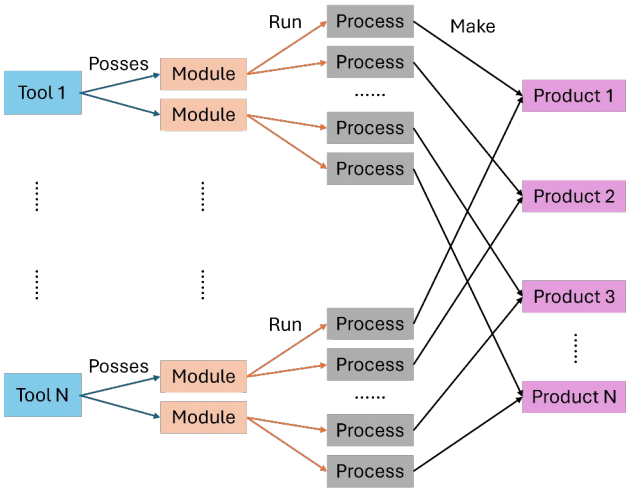

Figure 2.1:The industrial etching equipment's manufacturing system consists of multiple etching reactors, each equipped with several modules capable of running different processes.The production begins with pure silicon wafers, which undergo a series of processing steps before transform-ing into the final product.

goricalfeatures.Thestreamed,numericaldata,sourcedfromvariousphysicalsensors,include operational measurements such as pressure and gas flow, i.e., state variables that can be measured directly and in real-time.Categorical features describe process and material specifics that are de-cisions about the use of the machine.The two features included are an alphanumeric process ID thatcapturesthe"recipe"ofmachinefunctionsforaparticularproductandthesubstratefamily ID.Eachfamilyisafinitesetofclasses.Thesteamedoperationaldatawasaveragedforeach variable over duration of a batch run which is typically on the order of 20 minutes.Data collected cover five years of operation, from February 2018 to December 2022.The five etch tool datasets arecategorizedbymachinenumberandprocessmodule(singlemachinesthathaveparalleletch modules) as T51-PM1, T52-PM1, T52-PM2, T53-PM1, and T53-PM2.

Again, the solution objective is to build a binary prediction model by mapping these operating data to PASS and FAIL outcomes.Data from 2018 through 2021 are allocated for training,vali-dation and the tuning of the model hyperparameters.The 2022 data,the most recently collected,

were reserved for testing.Selecting data from the oldest rather than the newer sets and then testing withthemostrecentdataensuresthatmodeleffectivenesswithrespectto“normal”operational shifts and machine changes over time are accounted for and evaluated in the context of current and future production scenarios.

In addition to data selection, data preprocessing is also crucial for preparing datasets for model training process. Prior studies [21, 22] have highlighted that preprocessing can significantly impact model training performance, underscoring the importance of selecting those preprocessing meth-ods that optimize model performance for an operational situation[23]. In this work, preprocessing involves removing or padding invalid/null data points, encoding discrete variables, and normaliz-ing numerical data points for maximum performance accurately.The workflows for preprocessing inputs and outputs are graphically illustrated in Figure 2.2 (input data) and Figure 2.3 (outputs).

WithreferencetoFigure1,thefirststepisremoveinvaliddata.Aninvaliddatapointis defined when critical sensor data are absent, such as the output measurements of oxide thickness, the pass/fail status, or when a substantial number of the input sensor data is missing.Missing data points can be attributed to the temporary malfunction of sensors within a batch run, not uncommon with industrial data.In scenarios where extensive numbers of sensor data are missing (usually over half of the features), zeroing all missing features or feature paddings could detrimentally affect the dataset’sintegrityand,consequently,themodeltrainingperformance.Forsensordatapointsin which a small number of the measurements (usually only one or two features) are missing, these missing values are replaced by the average value of the remaining data points in the data set.This imputation method helps maintain the consistency and reliability of the dataset, ensuring that the training process is not skewed by gaps in the data. This strategy of handling missing data preserves the underlying statistical relationships and prevents the introduction of bias that could mislead the learningalgorithm.Wenotethatthesensordataiscollectedastime-seriesmeasurementsand averagedovertheduration ofeachprocessingsteptogenerateresentativevaluesforvarious features.Once this data is properly recorded,it is treated as accurate and reliable,as the sensors

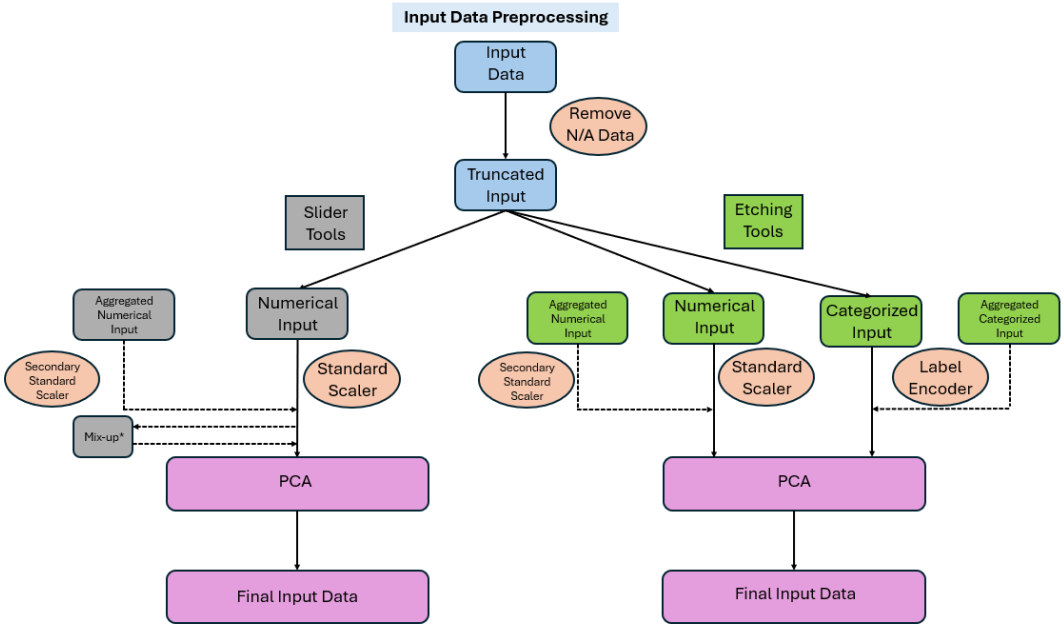

Figure2.2:Thisfigureillustratestheinputdatapreprocessingworkflowforetchingtoolsand slider production tools. Common steps include invalid data removal, normalization, and dimension reduction.Etchingtoolsincorporateencodersforcategoricaldatamanagement,whereasslider tools utilize Mixup to mitigate data scarcity.

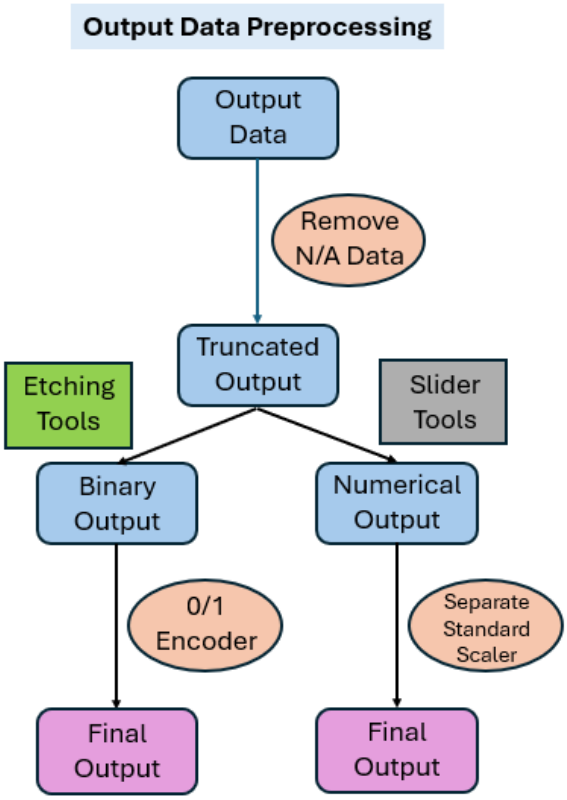

Figure2.3:Thisfigureillustratestheoutputdatapreprocessingworkflowforetchingtoolsand sliderproductiontools.Commonstepsincludeinvaliddataremoval,etchingtoolsrequires0/1 encoding, and slider tools requires separate scaling.

on the machine have been qualified by operations to ensure proper functionality and measurement accuracy.

Conversionofcategoricalfeaturesintonumericalformatsisessentialfortheirintegration into numeric-based machine learning models.Each categorical feature, such as the process ID and the substrate family ID, must be uniquely encoded using methods such as label encoding or one-hotencoding.Inthiswork,labelencodingwasusedtoconvertprocessnameIDsandsubstrate family IDs into numerical values.To ensure consistency across different machines, substrates, and production processes, the encoding was created using all available datasets combined, covering all possibleentries.Thisapproachpreventsissuesthatcouldariseiftheencoderweretrainedona limited dataset.For example,if a tool or module only runs certain processes,an encoder trained only on that data might fail when applied to another tool with different processes, leading to errors due to missing values.By including all possible entries during encoding, we ensure that the data remainscompatibleacrossdifferenttoolsandprocesses.Fortheoutputvariable,whichiseither PASS or FAIL condition, binary encoding scheme is adopted, where ‘1’ represents a PASS and ‘0’denotes a FAIL.

Normalization and scaling of data are critical for optimal training performance, when the input features span wide numeric ranges and scales.For instance, in this use case some features might rangefrom0to1,whileothersrangefrom0to100,000.Normalizationwascriticaltoavoiding gradient vanishing or gradient explosion [24].Although gradient-boosted, tree-based methods like XGBoostandAdaBoostareinherentlylesssensitivetoscaledifferencesduetotheirrelianceon decisiontreestructures,normalizationstillplaysasignificantrole.Unscaleddatacanintroduce inefficiencies in node splitting such that features with larger value ranges can dominate the feature selectioninsplitdecisions,potentiallyleadingtosuboptimalsplitsthatdonotcapturethetrue underlying patterns in the data effectively.

To ensure all features are normalized consistently across different datasets, a standard scaler is applied to each dataset independently, normalizing the features to a mean value of 0 and a standard

deviation of 1.This scaling process adheres to the formula:

$$Z = \frac{X - u}{s}$$

where Z is the output vector of a scaled numerical feature, X is the input vector of the original feature, u is the average value of X, and s is the standard deviation of X. In the context of data aggregation, a method detailed in a previous work [1], where model training involves combining multiple datasets to improve the model performance by increasing data variability and data volume, it is found to be beneficial to scale each dataset separately prior to concatenating datasets. In other words, the concatenations should happen between scaled datasets but not raw data. This method is necessary because, within each module's reaction chamber, the distribution of features often varies significantly. For example, although the same feature across different datasets may be similar in scale, direct scaling of the aggregated dataset could distort the inherent distributions of each dataset if the differences in scale and standard deviation are substantial. As shown in Ta-ble 2.1, the mean percentage difference of average values across all features exceeds 40% between most tools, indicating significant distribution shifts between datasets. This distribution distortion can mislead the model if the datasets are concatenated before normalization, then impacting its performance. Consequently, separate scaling for each dual-tool combination is implemented to maintain the integrity of their original distributions relative to the output. This approach preserves crucial relationships within the data, ensuring more reliable and accurate model training outcomes.

Table 2.1:Mean Percentage Difference of Average Value of All Features

	T51-PM1	**T51-PM2**	**T51-PM3**	**T52-PM2**	**T53-PM2**
T51-PM1	N/A	64%	14%	90%	43%
T51-PM2	77%	N/A	75%	55%	46%
T51-PM3	20%	86%	N/A	101%	66%
T52-PM2	5865%	3208%	5274%	N/A	9932%
T53-PM2	47%	45%	45%	81%	N/A

Afterthenormalizationandaggregationprocess,dimensionreductionisfacilitatedusing Principal Component Analysis (PCA). It is worth noting that PCA requires the data to be normalizedbeforeitsapplication.Normalizationstandardizestherangeoffeatures,ensuringthateach feature contributes equally to the analysis and that features with larger scales do not dominate the principal components.This prerequisite is crucial because PCA is sensitive to any variance in the initial variables.The PCA process begins by calculating the covariance matrix for the data, which identifies the directions in which the data varies the most.If the data are not normalized, features withhigherabsolutevaluescoulddisproportionatelyinfluencethecovariancematrix,leadingto biasedprincipalcomponentsthatdonotaccuratelyreflecttheunderlyingdatastructure.Subse-quently, the eigenvalues and eigenvectors of this covariance matrix are computed.The eigenvalues are sorted in descending order, and their corresponding eigenvectors are aligned accordingly.The principalcomponentsareselectedbasedonthesesortedeigenvalues,thosethatexplainthemost variancearetained(inthiswork99.9%ofvariancesareretained),discardingthelesssignifi-cant components (non-contributing features, noise).This selection is critical because features that exhibithighervariancearconsideredmoreinfluentialforthemodel'spredictiveaccuracy.The calculation process is shown below:

$$Q = X_T X = W \Lambda W_T \tag{2.2}$$

where$X$representsthedatamatrixwithdimension($k \times d$),where$k$denotesthenumberofdata points and d represents dimensions of datasets.Q refers to the covariance matrix, which is a square matrixofdimension($d \times d$).Λisadiagonalmatrixconsistingofeigenvaluesof$Q$arrangedin descendingorder.Correspondingly,Wisamatrixoftheeigenvectorsof$Q$,alignedinthesame sequenceastheeigenvaluesinΛ.Thelineartransformationfromtheoriginaldataspacetothe

principal components space is represented by equation below:

$$T_L = XW_L \tag{2.3}$$

where L is the number of reduced dimensions determined by the proportion of variance to be retained, T_L represents the transformed data matrix with dimension ($k \times L$), and W_L is the first L columns of W matrix.

Each principal component is a linear combination of the original features, representing a new direction in the feature space. By focusing on these principal components, PCA effectively reduces the dimensionality of the data. This reduction not only compresses the data with minimum loss by emphasizing directions with most significant variations, but also helps in mitigating the risk of overfitting by reducing feature numbers and noise in the data. In this work, 99.9% of variances are retained, and the resulting feature space has dimension between 10-15, corresponding to 60% to 75% reduction in dimension.

2.1.2 Slider Production Tools

Slider tools are a different part of the manufacturing process from wafer production where machine tools are used to make specific parts used in the production of data storage devices. For this study, we focused on a milling machine step which is used to establish a critical surface depth for the slider component. The modeling objective is the same virtual metrology solution for wafer produc-tion in that milling machine sensor data were collected, contextualized, processed and engineered for building a model to predict milling depth. In this study we have benchmarked data processing methods and workflow, and we compare and contrast the approach to building the virtual metrology solution for two different production applications. As with wafer production, modeling begins with data analysis. Since the goal is to predict milling depth, a continuous variable, a regression model is developed. This model leverages multiple features, approximately 20 in total, to estimate milling

depth with precision.However, an important observation from the data is that milling depth values tendtoclusterintodiscreteranges,correspondingtospecificproductspecifications.Figure2.4 illustratestheseclusters,withmillingdepths predominantlyfallingwithingroupsof1450–1550, 1650–1850,and2150–2350.Theseclustersreflectthelimitedvarietyofproducts,eachwithits own depth requirement, and are an inherent characteristic of the manufacturing process.

Given this natural grouping, the regression output can be easily mapped into predefined depth ranges,allowingforamulti-classclassificationapproach.Insteadofbuildingaseparateclassification model, the regression predictions are assigned to the nearest cluster, effectively classifying each wafer's milling depth into a corresponding product category.Additionally, this classification approach aids in identifying failures.Failures are defined as wafers whose milling depth deviates significantlyfromtheexpectedclustersforagiventool-modulecombination.Forexample,ifa particulartoolisdesignedtoproduceonlyonetypeofproduct,anywaferwithamillingdepth outside its designated range is considered a failure, even if its depth aligns with a valid range for anotherproduct.Inthiscontext,failuresindicatedeviationsinproductionquality,highlighting potential process anomalies.This depth-failure classification enhances both the model's practical utility and its ability to detect defects in real-world manufacturing.

For classification, data points outside of the major depth ranges are flagged as failures, with thedistributiondetailsoftheseprincipalrangescatalogedinTable2.2.Thedatasetsrelevantto sliderproduction,namelyT07-PM1,T07-PM2,T15-PM1,T02-PM1,T02-PM2,T01-PM1,and T05-PM2, consist of 20 features each, similar to the datasets from etching tools but encompassing fewerfeaturesanddatapoints.Theorganizationofdatapointswithinmajordepthcategories foreachdatasetisalsodocumentedinTable2.2,whererednumbersmeandatapointsinoutlier regions.

Data preprocessing for these datasets follows a similar approach to that used for etching tools, includingtheremovalofnullvalues,paddingofmissingdata,normalization,aggregation,and dimensionreduction.However,sincethesliderproductiondatasetsdonotcontaincategorical

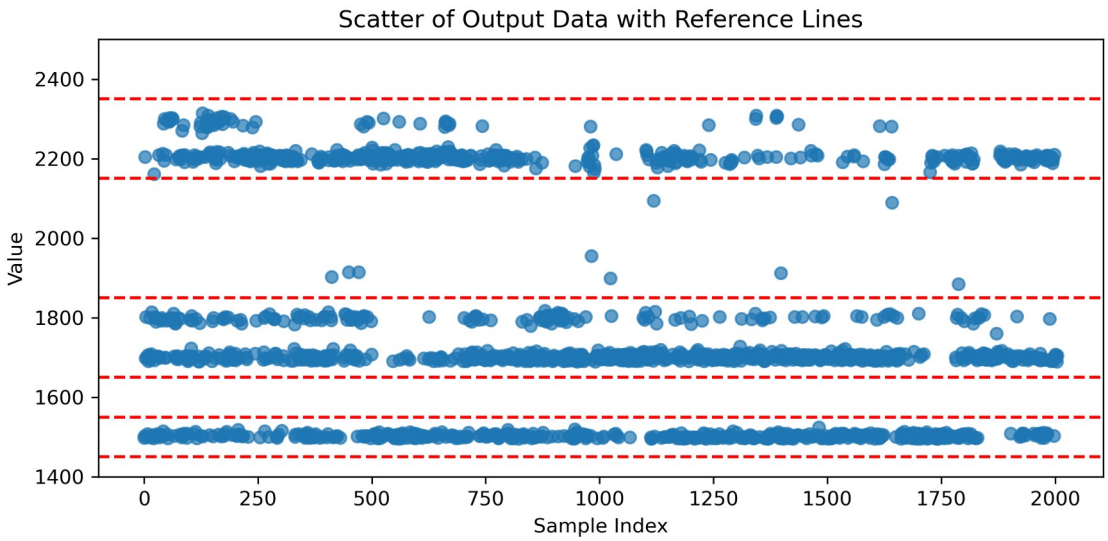

Figure 2.4: Scatter plot of T15-PM1 output, illustrating how data points are primarily grouped into several clusters.

Table 2.2:Data Distribution Across Different Ranges

	1450–1550	**1650–1850**	**2150–2350**	**Others**	**Total**
T07-PM1	1	869	9	22	901
T07-PM2	7	992	4	11	1014
T15-PM1	518	943	530	15	2006
T01-PM1	98	198	158	11	465
T02-PM1	0	742	108	24	874
T02-PM2	1	685	18	24	728
T05-PM2	0	153	0	3	156

features, the encoding step is not required. As shown in Figure 2.2, unique to slider tools, however, istheapplicationofseparatescalingontheoutputdataduringaggregation,astheoutputsof etching tools datasets are binary values.In this aggregation process,one dataset is designated as theprimaryor'major'datasetaimedatperformanceenhancement,whileotherdatasetsserveas supplementaryor'helper'datasets.Theoutputscaleristailoredtothedistributionofthemajor datasetbeforebeingappliedtothecombineddatasettopreserveitsintegrity.Thisstrategyis crucial because using a universal scaler across the aggregated data could distort the distribution of the major dataset of interest, such as the single-region T07-PM1 versus the three-region T15-PM1.

After normalization and aggregation, PCA is applied to the input data to retain at least 99.9% of the original variance, significantly reducing dimensionality while preserving essential informa-tion.The number of principal components retained varies by dataset, typically resulting in about 10 principal components, effectively halving the original feature set. This dimensionality reduction is critical in managing the complexity and enhancing the interpretability of the models developed from these high-variance datasets.

Duetothelimiteddataavailabilityforsliderproductiontoolsascharacterizedbyashorter datacollectionperiodofonlyoneyearcomparedtofiveyearsforetchingtoolsandalowerfre-quencyofdatarecording,enhancementsinmodelperformancearenecessary.Toaddressthis, adataaugmentationtechniqueknownasMixupwasemployedtocreateartificialdatapointsby interpolation between existing data points.

However,giventhatthesliderdatabelongstoseveraldiscreteregions,interpolatingdata points between major regions was avoided since doing so could lead to non-existing milling depth. Fordataaggregation,eachdatasetundergoesMixupindependentlywithinitsrespectivemajor regions,andtheresultingpost-processeddatasetsarethenconcatenated.Thisensuresthatthe augmented data remains representative of the true distribution patterns observed in the production environment.Toevaluatetheimpactofdifferentinterpolationstrategiesonmodeleffectiveness, two distinct Mixup schemes were tested.The first scheme is tri-point interpolation, which creates

two artificial data points between each two data points.This scheme is formulated as follows:

$$x_{mix} = 0.33x_i + 0.67x_{i+1}, x_{mix} = 0.67x_i + 0.33x_{i+1}$$
$$y_{mix} = 0.33y_i + 0.67y_{i+1}, y_{mix} = 0.67y_i + 0.33y_{i+1} \quad (2.4)$$

The second scheme creates one data point directly in the middle of two data points as formulated as follows:

$$x_{mix} = 0.5x_i + 0.5x_{i+1}$$
$$y_{mix} = 0.5y_i + 0.5y_{i+1} \quad ($$

Beyond data augmentation, Mixup also acts as a form of regularization, effectively smoothing thedecisionboundariesofthemodel.Thissmoothingisachievedbyencouragingthemodelto perform linear interpolations between features and their associated targets in the input space, which can reduce the model's confidence in far-reaching predictions.Such a characteristic is particularly useful in mitigating overfitting, as it prevents the model from learning overly complex patterns that areheavilydependentonthespecifictrainingdatadistribution.Instead,Mixupencouragesthe model to generalize better to new,unseen data by promoting a broader exploration of the feature spaceandreducingthelikelihoodofdrasticoutputchangesinresponsetosmallvariationsin input.ThisregularizationeffectmakesMixupavaluabletechniqueinenhancingtherobustness and generalizability of machine learning models [20].

2.2 Model Training

2.2.1 Introduction to Machine Learning Models

FeedforwardNeuralNetworks(FNNs) arewidely usedformodeling processesinvolving regressionandclassificationunderlargeamountsofdataandfeatures.However,tree-basedensemble methodslikeExtremeGradientBoosting(XGBoost)demonstrateadvantageswhendealingwith

high-dimensional feature spaces and significant noise levels.This is because:

- •FNNs require careful normalization,can consume more computational resources,and may suffer from overfitting in complex architectures.
- •Tree-based approaches (like XGBoost) require less stringent normalization, often train faster on moderate size dataset,and are generally robust to overfitting due to inherent regulariza-tion.

In industrial applications, datasets often present challenges such as high-dimensional feature spaces,missingvalues,classimbalance,andthepresenceofsignificantnoise.Thesecharacter-isticsmaketree-basedapproachesparticularlywell-suitedforindustrialdatamodeling.Thekey advantages include:

- •**RobustnesstoNoisyandIncompleteData:**Industrialdatasetsfrequentlycontainsensor readings,production metrics,and operational parameters that may be affected by measure-menterrors,environmentalconditions,ormissingentries.Tree-basedapproacheshandles such inconsistencies effectively using its optimized split criteria and missing value handling mechanisms, allowing it to make robust predictions despite imperfect data.
- •**Feature Selection and Interpretability:**Unlike FNNs, which often require careful feature engineering and are perceived as black-box models, tree-based approaches naturally identi-fies the most important features during training.This ability is critical in industrial settings whereengineersanddecision-makersneedinterpretableinsightsforprocessoptimization, fault diagnosis, and predictive maintenance.
- •**ComputationalE**ffi**ciencyandScalability:**Industrialdatasetsinthisstudyrangefrom 10,000 to 30,000 data points.Training deep neural networks on such data is more compu-tationally expensive than XGBoost, even with a GPU, and require extensive hyperparameter tuning.**Tree-based approaches, on the other hand, is optimized for pseed and scalability for**

mid size datasets as in this work, leveraging parallel processing and tree-pruning techniques to efficiently handle large datasets with minimal computational overhead.

- **RegularizationforGeneralization:** FNNs,particularlydeeparchitectures,requirecare-fultuningofdropoutratesandbatchnormalizationtopreventoverfitting.Tree-basedap-proachescanincorporatebuilt-inregularizationtechniquessuchasL1(LASSO)andL2 (Ridge) penalties, along with shrinkage (learning rate adjustment), ensuring better general-ization performance without extensive fine-tuning.

Giventheseadvantages,tree-basedapproachesemergesasahighlyeffectivetoolformod-elingindustrialprocesses,whererobustness,interpretability,andcomputationalefficiencyare paramount.XGBoostisoneexampleofatree-basedensemblemethodcombiningthepredic-tions of multiple decision trees in a gradient-boosted framework.The algorithm begins by training a base tree, then computes the residual (the difference between the current prediction and the true label).Subsequent trees are trained to optimize these residuals.The final prediction for sample x_i after t trees is:

$$\hat{y}_{(t)\,i} = \hat{y}_{(t-1)} + \eta \cdot f_t(x_i), \quad ($$

where η is the learning rate and f_t is the t-th decision tree. This iterative approach refines the model incrementally,makingXGBoostparticularlyeffectiveinmanystructureddatatasks,whichare tasks thatinvolve datasetswhere features are well-defined,typically organized in tabular format, and consist of numerical,categorical,or ordinal variables.These tasks are common in industrial settings, where data is collected from sensors, production logs, and quality control systems.

2.2.2 Model Training on Etching and Slider Tools

Etch tool data are used to train a ML classification model to perform a binary prediction of PASS or FAIL product at the completion of the machine tool process step.Because the FAIL rate from the

process step is small(~2%), the data are highly imbalanced between PASS and FAIL. We therefore useaweightedcrossentropylossapproachtoemphasizeFAILconditions.Cross-entropyloss, also known as log loss, is a commonly used loss function for classification tasks.It measures the divergence between the true labels and the predicted probabilities, penalizing incorrect predictions more heavily.In binary classification, it is defined as:

$$L(y_i, \hat{y}_i) = -w_0(1 - y_i)\log(1 - \hat{y}_i) - w_1 y_i \log(\hat{y}_i), \tag{2.7}$$

where w_0 and w_1 are class weights set as:

$$w_1 = 1 \; n_1 \quad ,$$

and n_0 and n_1 denote the number of PASS and FAIL samples, respectively. It's worth mention-ing that in our datasets FAILs are usually represented as 1 instead of 0.This scheme emphasizes the minority class to improve the detection of FAIL events.

80% of the etch tooling dataset (2018–2021) was used for training, while the remaining 20% was separated out for validation.To systematically tune hyperparameters of the machine learning model, we performed a 5-fold cross-validation on the training data.Cross-validation is a technique usedtoassessthegeneralizabilityofamodelbysplittingthedatasetintomultiplesubsets.In 5-foldcross-validation,thetrainingsetisdividedintofiveequallysizedfolds,wherethemodel istrainedonfourfoldsandvalidatedontheremainingfold.Thisprocessisrepeatedfivetimes, with each fold serving as the validation set once.The final model performance is averaged over all iterations, reducing the risk of overfitting and improving robustness.

To further optimize hyperparameters, we BM quantum computers leverage furniture a grid search exploring various learning rates, thenumberofestimators,andregularizationstrengthsforbothFNNandXGBoost.Gridsearch systematically evaluates predefined combinations of hyperparameter values to identify the optimal setthatmaximizesmodelperformance.Thisapproachensuresthatdifferentconfigurationsare

testedcomprehensively,balancingmodelcomplexityandpredictiveaccuracy.Thefinalselected hyperparameters are shown in table 2.3 and table 2.4, respectively, where the underlined parameters represent the chosen optimal values.

Table 2.3:Hyperparameters for FNN

Hyperparameter	**Range/Candidate Values**
Number of layers	[1, 2, 3]
Number of neurons	[16, 32, 64]
Activation function	[’relu’, ’sigmoid’]
Dropout ratio	[0.0, 0.2, 0.5]
L2 regularization	[0, 10^{-4}, 10^{-3}, 10^{-2}]
Training epochs	[500, 1000, 2000]
Early stops	[’Yes’, ’No’]

Table 2.4:Hyperparameters for XGBoost

Hyperparameter	**Range/Candidate Values**
Learning Rate	[0.1, 0.3, 0.5]
Max Tree Depth	[6, 8, MAX*]
Subsample Rate	[0, 0.5, 1]
L1 Regularization	[0, 1, 10]
L2 Regularization	[0, 1, 10, 100]

* Dimension after PCA

In all kinds of machine operation that run continuously over a long periods of time,process drift poses a significant challenge to maintain accurate model performance over time, as even the sensors themselves can drift and develop bias.Process drift occurs due to various factors such as equipment aging, accumulation of unwanted deposits on machinery, and other operational changes that alter system behavior with time.For example, in etch machines, process drift can manifest as

a gradual reduction in etch rate due to the accumulation of impurities on chamber walls[25].This buildup alters plasma conditions, leading to deviations in feature dimensions and film thicknesses overtime.Ifnotaccountedfor,suchdriftcandegradethepredictiveperformanceofmachine learning models trained on historical data, necessitating periodic model recalibration.

As a result, models trained on initial datasets may gradually lose accuracy.Transfer learning addressesthisbyadaptingexistingmodelstonewdatawithoutfullretraining,preservingprior knowledgewhileintegratingnewinformation.Thisisespeciallyvaluableinindustrialsettings, wherecollectingsufficientdataindifferentenvironmentsiscostlyduetotheneedforphysical measurements.Additionally, in scenarios such as data drift, continuous updates are required, but new data may never be sufficient for training from scratch before being outdated.

Our implementation of transfer learning followed a three-step procedure designed to improve model performance while minimizing the need for costly physical measurements:

1. **Base Model Training:** The procedure begins with developing a base model by training the feedforwardneuralnetwork(FNN)onolddata,whichinourcaseisthedatafrom2018-2021, to establish foundational performance metrics.This base model serves as the starting point for all subsequent updates and evaluations.

2. **IncrementalModelUpdatingwithSGD:** Oncethebasemodelisestablished,itisincrementallyretrainedonnewdatausingstochasticgradientdescent(SGD).Thetrainingdata windowsize(i.e.,mini-batchsize)isatunablehyperparameter,withexamplesincluding mini-batches of 1 or 10 data points, referred to as ’train data length’ inTable 2.5.To prevent dataleakage,eachnewdatapointisfirstevaluatedusingtheoriginal,pre-updatedmodel before being used for training.Although the true label is available, the model is tested as if it had not yet seen this data point.Only after recording this evaluation is the model updated with the new data using SGD. This ensures an unbiased assessment of how well the model adaptsovertimewhileaccuratelymeasuringtheeffectivenessoftheupdatingalgorithm.

The use of SGD with a carefully selected learning rate enables efficient updates,striking a balance between adapting to new information and preserving existing knowledge to prevent model destabilization.

3.**Model Evaluation on Test Data:**After the model is updated in the previous stage, it is now evaluated on new test data points.At this stage, the model is not updated—only its performance is measured.The ratio between the number of test data points and the training data points from the previous step is defined as the ’Test/Train Length Ratio’ inTable 2.5.This evaluation step ensures that the model can make accurate predictions on new data without ad-ditional retraining. The goal of transfer learning here is to improve model performance while minimizingtheneedforphysicalmeasurements,whicharecostlyandtime-consuming.If themodelwereretrainedoneverynewdatapoint,itwouldrequirecollectingnewlabels throughphysicalmeasurements,defeatingthepurposeoftransferlearning.Althoughall datainthisworkarelabeledthroughphysicalmeasurements,thelabelsareusedsolelyto evaluatetheeffectivenessoftransferlearning.Fromthemodel’sperspective,itdoesnot have prior knowledge of the labels during training.By testing the transfer learning scheme in this controlled setting, its effectiveness can be validated before future real-world applications.After completing this evaluation step, the process returns to the previous stage, where the model is updated with the next batch of new data points, continuing the learning cycle. After completing Step 3, the process returns to Step 2, where the model is updated with the next batch of new data points.This continuous cycle of evaluation and updating is known as real-time update, allowing the model to adapt dynamically to evolving data patterns.

Hyperparameterssuchasthelearningrate,trainingdatalength,andtest/trainlengthratio areoptimizedtoachieverobustmodelperformance.Thisoptimizationisconductedthrougha systematic grid search, as detailed in Table 2.5.For each tool, all combinations of train data length and train/test length ratio are thoroughly tested to examine the effectiveness of transfer learning.

Table 2.5:Hyperparameters for Transfer Learning Model Training

Parameter	**Hyperparameters**
Learning Rate	[0.0001, 0.001, 0.01] [1, 10]
Training Data Length Test/Train Length Ratio	[1, 2, 4]

Insummary,transferlearningefficientlymaintainsmodelaccuracyamidprocessdriftby leveraging knowledge from historical data and incorporating incremental updates.This approach sustains performance while reducing the costs of full retraining, making it well suited for evolving industrial processes and the practical constraints of manufacturing environments.

For the slider datasets, the task is a regression problem instead of binary classification. There-fore, the Mean Squared Error (MSE) is employed as the primary loss function:

$$N \quad \sum_{i=1}^{N} \quad \hat{y}_i - y_i \quad 2, \qquad (2$$

where$\hat{y}_i$and$y_i$arethepredictedandactualvaluesforsample$i$,and$N$isthetotalnumberof samples.

Duetothelimiteddatasetsize(collectedoverashortertimeperiod),fewerlayersandneu-ronsareusedintheFNNtopreventoverfitting.XGBoost,however,retainsitsstandardsetof hyperparameters, exploiting its tree-based mechanism to handle even relatively small datasets.To prevent overly small partitions, we use an 80%–20% train/validation split without a separate time-based test set.Given the limited data volume and its confinement to a one-year range, time-based prediction is less critical.

ByapplyingFNNandXGBoostwiththesetailoredhyperparametersandtheMSEloss,we achievereliableregressionperformancewithintheconstraintsofasmallerdataset.Giventhe limited data, the FNN architecture is kept simple with fewer neurons to avoid overfitting and reduce computational costs.

Table 2.6:Hyperparameters for FNN

Hyperparameter	**Range/Candidate Values**
Number of layers	[1*, 2, 3]
Number of neurons	[8, 16, 32]
Activation function	[’relu’, ’tanh’]
Dropout ratio	[0.0, 0.2, 0.5]
L2 regularization	[0, 10^{-4}, 10^{-3}, 10^{-2}]
Training epochs	[500, 1000, 2000]
Early stops	[’Yes’, ’No’]

* Single Training

Chapter 3

Results and Discussion

3.1 Etching Tools

Theperformanceofthemachinelearningmodelsintroducedin Chapter2.2.2thatweretrained on industrial data we received are best evaluated within the bounds of the specific machines,operational environments, and conditions in which they will be applied.As discussed in a previous study,theetchmachinesinthisstudyperformwellwithlowpercentagesofproductFAILrates; therefore,thedatadistributionishighlyimbalancedtowardPASSvs.FAILruns[1].Thereare four possible outcomes in a binary classification:

1.**True Positive**:Correctly identifying a PASS.

2.**False Negative**:Incorrectly labeling a PASS as a FAIL.

3.**True Negative**:Accurately detecting a FAIL.

4.**False Positive**:Incorrectly labeling a FAIL as a PASS.

True positives and true negatives indicate that the classification model is correctly qualifying the product state.False negatives,though undesirable,can be mitigated since all flagged failures

undergoanadditionalmetrologyinspectionwherefalsenegativescanbecorrectedbymanual measurements.The larger concern in these operations are false positives, as they are a much more expensive miscategorization compared to false negatives.False positives are more costly because downstream operational resources continue to process the defective product, wasting resources and time.In most cases,false positives will eventually be identified through later metrology steps or final product reliability testing [26].

An acceptable classification model for this operating situation needs to have sufficient accuracy and precision in identifying true FAIL and PASS results but also in minimizing false positives to reduce the cost of classification errors. In operational terms, this virtual metrology solution leads to significant economic benefit by debottlenecking the physical metrology step and increasing pro-duction volume.

Theperformancemetricsfortheclassificationmodelsmustreflecttheabilityofamodelto control its false positive rate.Performance is adjusted by modifying classification thresholds that affect the balance among all four possible classifications. Increasing the sensitivity to misclassified wafers reduces false positives but raises false negatives and vice versa.For this application there is a need to demonstrate the ability to tune the model by considering performance across a range of threshold settings.Confusion matrices,while commonly used to evaluate classifier performance, only address results at a specific threshold[27].They fail to capture overall model performance for various settings.Accuracy evaluation is unsuitable because of the extreme PASS vs.FAIL im-balance.For instance, a model that always predicts "PASS" may appears highly accurate because the vast majority of classifications are PASS, but is functionally useless with a 100% false positive rate.

Eachmodeldoesnothaveasingle,fixedfalsepositiverate;rather,itsperformancecanbe adjusted by modifying classification thresholds.Increasing sensitivity to misprocessed wafers re-duces false positives but raises false negatives, whereas a more lenient approach does the opposite. Confusion matrices,while commonly used to evaluate classifier performance,only depict results

ataspecificthresholdandfailtocaptureoverallmodelcapabilityacrossvarioussettings.Other common evaluation metrics, like accuracy, are also unsuitable due to the extreme class imbalance in manufacturing datasets.For instance, a model that always predicts "PASS" may appear highly accurate but is functionally useless with a 100% false positive rate.

TheReceiverOperatingCharacteristic(ROC)analysisisabetteranalysistoolforthisuse case because it directly plots the true positive rate (TPR) and false positive rate (FPR) across the entire span of thresholds.The output of the classifier produces continuous scores in the range [0,1] of the classification threshold.The default threshold is of 0.5, which is equivalent to guessing an outcome.We therefore want to be able to adjust this threshold to get an acceptable performance. Setting the threshold to 0 results in a TPR of 100% and an FPR of 100% (always PASS), while a threshold set at 1 results in TPR and FPR rates of 0% (always FAIL). The optimal threshold there-fore requires balancing sensitivity (TPR) angainst false alarms (FPR). To objectively compare the performance of models, the Area Under the ROC Curve (AUC) is used since it is a metric that quan-tifies overall performance across all threshold settings when plotting TPR against FPR. A perfect model would achieve an AUC of 1, meaning 100% sensitivity with no false positives.In contrast, amodelmakingrandompredictionswouldyieldanAUCof0.5.Sincemanufacturingdatais typically imbalanced, ROC-AUC provides a reliable measure of classification effectiveness [28].

3.1.1 Single and Dual Tool Training

The AUC scores of the FNN and XGBoost models on each of the five etching machine datasets with single-tool(trainedonlyononedataset)anddual-tool(trainedontheaggregationoftwounique datasets, the aggregated dataset is picked by best available score) training are shown in Figure 3.1:

For single-tool training, we observed an increase in AUC scores for 3 out of the 5 tools when using the XGBoost model compared to the FNN model. The largest increase was 0.09 in

T53-PM1, whilethelargestdecreasewas0.13inT52-PM2.Indual-tooltraining,XGBoostdemonstrated

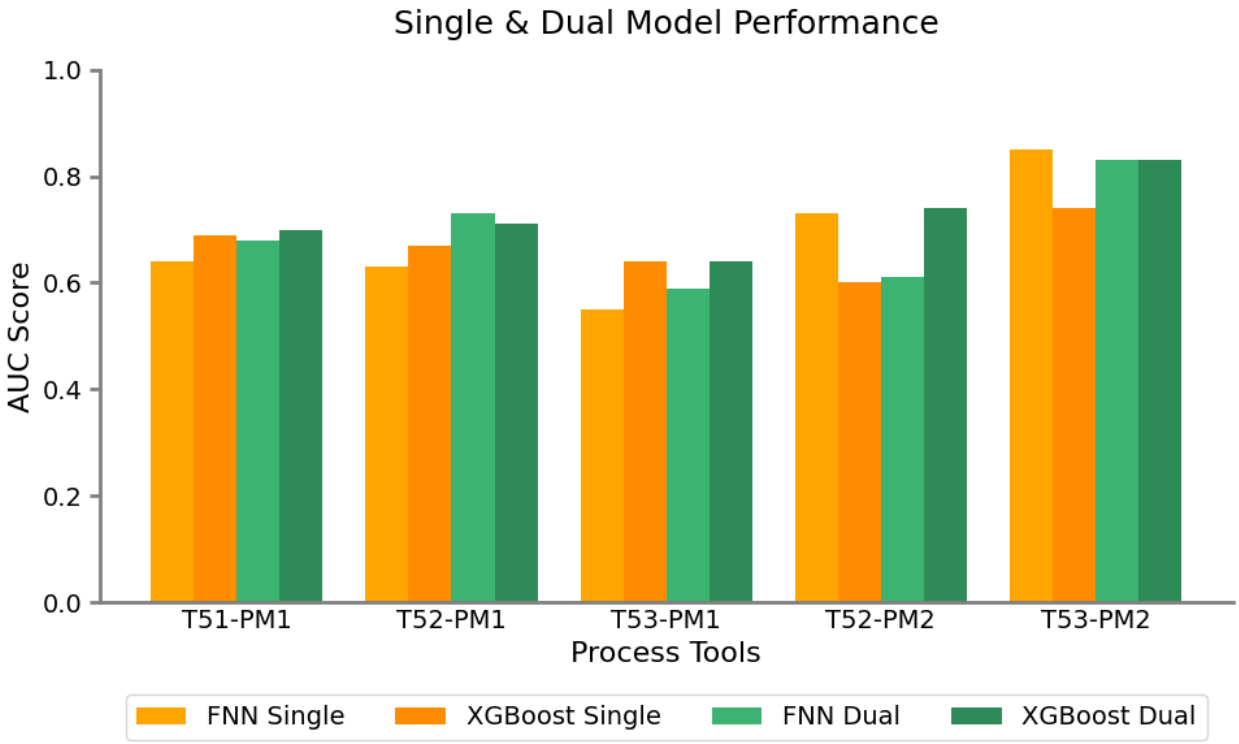

Figure 3.1: This figure shows the AUC scores for the FNN and XGBoost models applied to etching tool data using single tool and dual tool training.

betterorsimilarperformancecomparedtoFNNacrossalltools.ThelargestAUCincreasewas 0.13 in T52-PM2 and 0.05 in T53-PM1,with the largest decrease being minimal at 0.02 in T52-PM1.AnimprovementintheROC-AUCscoreindicatesabetterabilitytodistinguishbetween PASSandFAILclassifications.Forinstance,anAUCincreaseof0.13cansuggestthatunder controlled80%truepositiverate(TPR)thefalsepositiverate(FPR)hasdecreasedatleast10% (the actual level depends on the specific behavior, the AUC score does not have a fixed relationship withTPR/FPRimprovement)atvariousclassificationthresholds.Inpracticalterms,thismeans that XGBoost reduces misclassifications that fewer failed cases are incorrectly predicted as pass, or more true failures are correctly identified. Conversely, the AUC drop of 0.13 in T52-PM2 indicates a decline in this ability, likely due to the dataset's small size. To illustrate the practical implications, anAUCscoreof0.8onT53-PM2typicallyenablesdetectionofover90%ofdefectiveproducts (i.e., fulfilling the <10% FPR requirement) and reduces more than half of the associated physical measurementsneededforfurtherinspection(Falsenegativepredictions).Ingeneral,every0.1 incrementintheROC-AUCscoreyieldsapproximately10%reductionintherequiredphysical measurements.However, these improvements are not uniformly distributed; gains realized closer to the ideal AUC of 1.0 tend to be both more significant and more difficult to achieve.

Despitesomedecreasesinsingle-tooltraining,thereductionswerenegligibleindual-tool training,andAUCscoreseitherimprovedorremainedstableforalldatasets.Thisconfirmsthat XGBoost is a competitive classifier for PASS/FAIL classification in industrial datasets. The notable fluctuations in T52-PM2 are likely due to its limited dataset size—only 863 data points compared to over 10,000 in other datasets.This small sample results in a very limited test set with just 5 fail data points, introducing bias and randomness that undermine the reliability of performance metrics forthistool.Furthermore,aggregatingdataacrosstoolscontinuestobevaluable.Thedual-tool training advantage observed in prior work with FNN remains valid for XGBoost, with dual-training AUCscoresconsistentlyoutperformingsingle-tooltrainingacrossallfivetools.Coupledwith its superior or comparable performance, significantly lower computational resource requirements,

and easier hyperparameter tuning compared to FNN, XGBoost, along with other boosted decision tree models, emerges as a highly competitive candidate for PASS/FAIL classification in industrial applications.

3.1.2 Transfer Learning

The transfer learning results for three different train/test ratios in single-tool training are presented in Figure 3.2.As stated in Chapter 2.2, the train/test ratios (row) indicates the proportion of data points used for model updates within a given time period, the train data length (column) indicates thefrequencyofmodelupdating.TheheatmapsforeachtoolaregeneratedbyplottingAUC scores for each Test/Train Ration and Train Data Length pair.A darker shade of blue indicates a higher AUC score, meaning better model performance.The heat map indicates that in most cases, variations in the train/test ratio, ranging from 1:1 to 1:4, do not significantly affect the AUC score when the train data length remains constant.The performance differences between frequent model updates (small train/test ratio) and less frequent updates (large train/test ratio) are minimal.This indicates that the model does not require persistent updates to maintain its predictive accuracy.It can be trained on a subset of the data at periodic intervals and then applied over extended periods withoutsubstantialperformancedegradation.Thisfinding haspracticalimplications.Insettings where labeled data collection requires physical measurements, a reduced train/test ratio translates tofewernecessarymeasurementsinagiventimeperiod.Forexample,ina1:4train/testratio scenario, only 20% of the available data is used for model updates within an update period.This impliesthatifphysicalmeasurementsarerequiredforlabeling,themeasurementworkloadcan be significantly reduced, as labeling is only needed during model training.In contrast, during the prediction phase, which accounts for 80% of the time, no physical measurements are required.In addition, in semiconductor manufacturing, where data collection is automated and continuous, an important takeaway is that effective model updates do not require all collected data.Instead, only asmallsubsetissufficienttomaintainoptimalmodelperformancewithinagiventimeperiod.

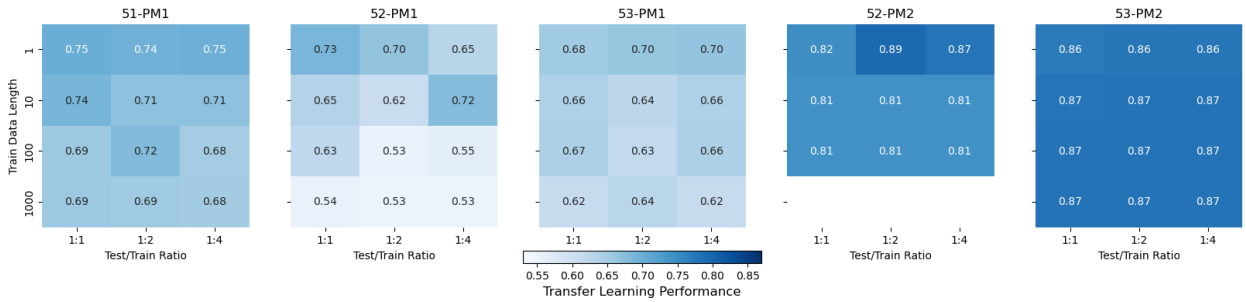

Figure 3.2:Single tool transfer learning results in heat map form.The performance difference in thesamerowisminimal(differenttrain/testlengthratio),buttheperformancedifferenceinthe same column (different train data length) is significant.

This insight allows for optimizing data usage, focusing computational resources on periodic model retraining rather than continuously processing every newly collected data point.

However,whencomparingdifferenttrainingdatalengths,acleartrendemerges:models trained with shorter datasets, which require more frequent updates in a time-series context, consis-tently outperform those with longer training datasets.This suggests that frequent adaptation of the model to recent data improves performance, likely because it enables the model to better capture the dynamics of the evolving system and mitigate temporal variations in the data distribution.In contrast, when the training data length is large, the model updates occur less frequently, potentially making it less responsive to shifts in the underlying data patterns.In particular, when the training data length is set to 1000 (excluding T52-PM2, which lacks sufficient data points), the performance closely resembles that of regular single tool training, indicating that transfer learning provides little to no benefit.In conclusion, the model should be updated frequently but in a low-density manner toimprovetheperformanceunderprocessdriftwithmaximumsavingscomputationalresources for retraining the model.

The results of transfer learning for dual tool training are presented in Figure 3.3 in the form of a heat map.Overall, the trends observed in single-tool training are largely retained, and while dual-tool training leads to slight performance improvements in certain cases, such as for T52-PM1

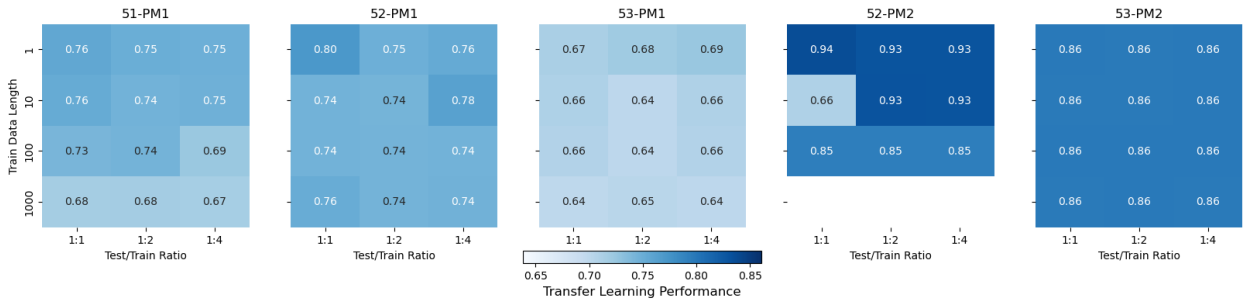

Figure3.3:Dualtooltransferlearningresultsinheatmapform.Theperformanceshowsslight improvements compared to single-tool training and follows similar trends observed in single-tool training.

andT52-PM2,theresultsformostothertoolsremaincomparabletosingle-tooltraining.This suggests that although data aggregation can enhance model robustness by exposing it to a broader range of conditions, its impact under transfer learning is limited.

Onekeyreasonforthisisthattransferlearninginherentlyallowsthemodeltodynamically adjusttoshiftsinthedatadistributionovertime.Thiscontinuousadaptationreducestheeffec-tivenessofthegeneralizationfromaggregatingbecausethemodelisalreadyevolvingtocapture new patterns.Additionally, while data aggregation improves generalization across different tools, transferlearning,particularlythroughfine-tuning,hastheoppositeeffectbymakingthemodel morespecializedforaspecifictoolonthemostrecentdataset.Asaresult,thebenefitofmodel generalizationduetoaggregateddatamaybeoverriddenbythemodel'sprogressiveadaptation torecentdata.Furthermore,transferlearningitselftendstobebiasedtowardsrecentdata,asit incrementallymodifiesthemodeltofitthelatestavailableinformation.Systematicapplication oftransferlearningtothemodelisarguablythemostsignificantreasonwhydataaggregationis lesseffective.Inshort,sincethemodeliscontinuouslyoptimizedforthemostrecentdataset,the contribution of earlier aggregated data diminishes over time.Consequently, while dual-tool train-ing offers advantages in other cases, the overall benefits of data aggregation in a transfer learning framework remains negligible.

3.2 Slider Tools

TheFNNandXGBoostregressionmodelsfor the sevenslider milling toolsintroduced in Chap-ter 2.2.2 are evaluated with two metrics:Median Absolute Error (MAE), measures the magnitude of prediction error,and median value is applied to remove the influence from outliers;and Coef-ficient of determination score (R^2), evaluates how model explains the variance of original dataset. Minimizing MAE and maximizing $R^2$are essential for improving model performance.

TheMedianAbsoluteError(MAE)measurestheaverageabsolutedifferencebetweenthe predicted and actual values.It is mathematically defined as:

$$\text{MAE} = MED(|y_i - \hat{y}_i|)$$

where$y_i$isthetruevaluevectorand$\hat{y}_i$isthepredictedvaluevector,MEDistheoperatorthat calculatesthemedianvalueofavector.Minimizingmedianabsoluteerrorisdesirablebecause, unlikemeanabsoluteerror,themedianislessaffectedbylargedeviations.Also,MAEcanbe beneficial over Mean Squared Error (MSE) because it is more robust to outliers, as it measures the mediandeviationratherthansquaringallerrors,whichreducestheimpactofextremevalues.A smaller MAE indicates that the model's predictions are closer to the actual values on average.

TheR^2score,alsoknownasthecoefficientofdetermination,indicateshowwellamodel explains the variance in the target variable.It is calculated using the following formula:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Here,$\bar{y}$ represents the mean of the actual values.The numerator is the sum of the squared errors, while the denominator is the total variance in the data.A higher R^2score, approaching 1, suggests thatthemodelaccountsforasignificantportionofthedata's varianceandithasastrongfit.R^2 isoftenusedtocomparetheperformanceofdifferentmodels,whereapositiveR^2impliesthat

the model performs better than a model that simply predicts the mean value.Achieving a high R^2 scorealoneisnotsufficient,becausetheremaystillbelargeindividualpredictionerrors,orthe model may overfit the data.Cross-validation is critical for confirming generalization capability of the model.

BalancingMAEandR^2ensuresthatthemodelmaintainsbothlowerrormagnitudesand strong variance explanation.A model with low MAE and a low R^2may fail to capture underlying patternsinthedata,whileamodelwithhighR^2butlargeMAEcansufferfromoverfittingor prediction instability.

3.2.1 Single and Dual Tool Training

Both the FNN and XGBoost models are trained similarly to the etch tools:single training, where training data only comes from the tool it is being tested on, and dual training, where training data comes from the tested tool and one of the other seven slider tools.Since the data range is primarily between 1450 and 2350,the goal is to ensure predictions remain within the controlled range and do not shift to another group.Therefore, an error below 20(~1%) is considered small enough for effectivemulti-classclassification,andanerrorbelow10(approximately0.5%)isregardedasa near-perfectscore.Intheslider productiontool,themeanabsoluteerror(MAE)maybeloosely comparedtothatofanetchingtoolbytreatingpredictionswithinthespecifiedmajorrangesas positive (true) and those outside as negative (false).Since this is a direct regression model, it is not feasibletoassignmultiplethresholdsforclassification;consequently,onlyasinglepairofTPR and FPR can be produced, permitting indirect comparison with ROC results.In general, an MAE ofaround20correspondstoanAUCofapproximately0.70–0.75,whereasanMAEbelow10 indicates an AUC exceeding 0.85.In this work, the R^2score is less critical than the absolute error, as the primary goal is to accurately predict the group to which each data point belongs.However, R^2remainsausefulmetricforassessinghowwellthemodelcapturestheoverallvarianceinthe dataset, providing insight into its explanatory power. The performances of both FNN and XGBoost

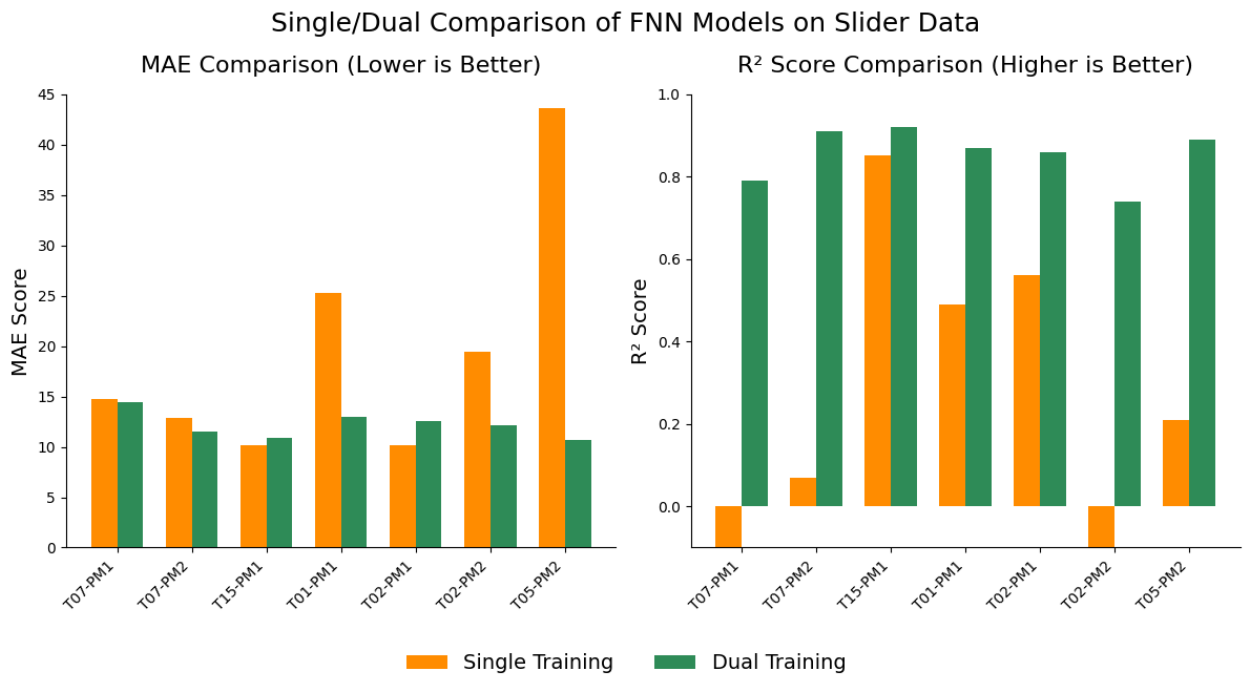

Figure3.4:ThisfigureshowstheMAEandR^2scoresfortheFNNmodelappliedtoslidertool data.MAE generally decreases when training data is aggregated with another tool.R^2generally increase with this aggregation as well.

models using single and dual training are shown in Figure 3.4 and Figure 3.5.

For the FNN model,MAE decreased for 5 tools and increased for 2 tools after aggregation. The largest decrease was 32.85 (~2%), while the largest increase was 2.36 (~0.1%).In general, the decreases in MAE were more significant than the increases.R^2scores increased for all 7 tools in the FNN model,with the largest increase being 1.37,change from a score below baseline to a score over 0.8.

FortheXGBmodel,MAEdecreasedfor5toolsandincreasedfor2tools.Thedecreasein MAEisnotaslargeasseenintheFNNmodelbuttheincreasesinMAEwaslessthantheone seen in the FNN model.The largest decrease in MAE was 3.86(~0.2%) while the largest increase was 1.54(~0.07%).Significant increase in R^2scores are also seen in the XGB model.R^2scores increased for 6 tools, the largest increase being 0.47 while the only decrease being -0.03. Although

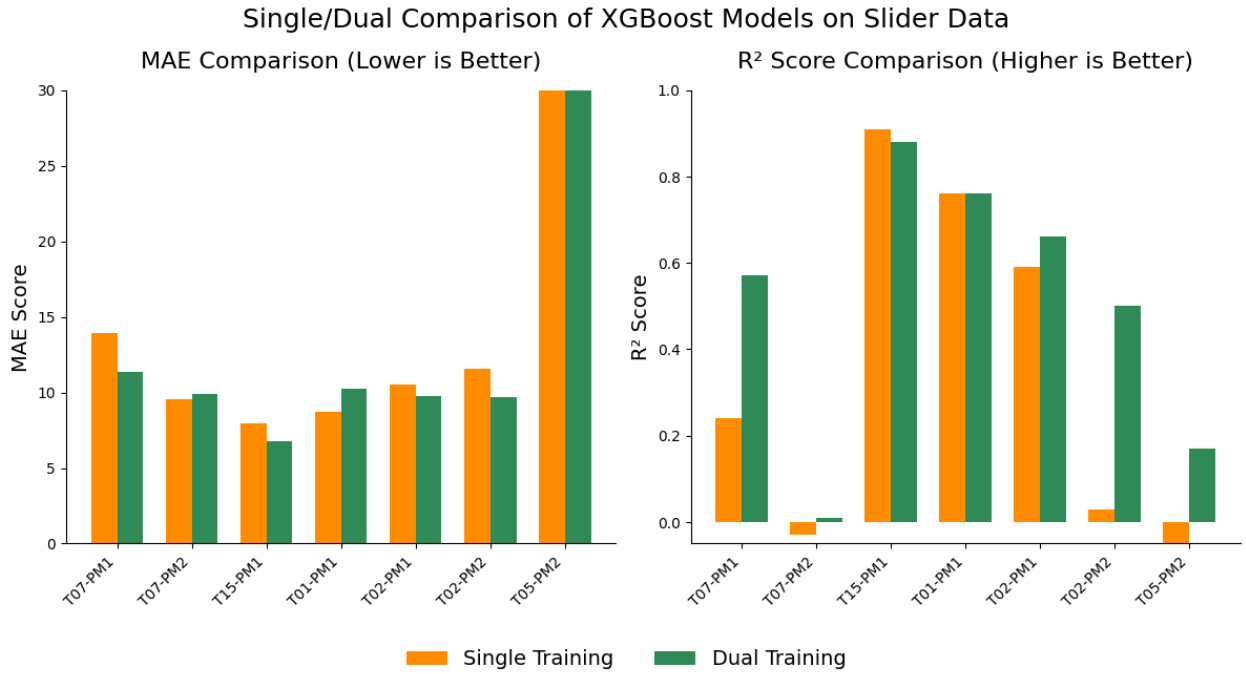

Figure3.5:ThisfigureshowstheMAEandR^2scoresfortheXGBoostmodelappliedtoslider tool data.MAE generally decreases when training data is aggregated with another tool, but not as much as seen in the FNN model.R^2generally increase with this aggregation as well.

the decrease in MAE from dual-tool training is small for XGBoost model, the MAE score predicted by XGBoost is lower for 6 out of 7 tools than FNN.

Theresultsdemonstratethataggregatingtrainingdatafromanothertoolimprovestheper-formanceofbothmodelsintermsofreducingMAEandincreasingR^2scores.However,the magnitude of improvement varies between the two models, with FNN showing more pronounced changes in performance metrics.This indicates that FNN benefits significantly from data augmen-tationinmostcases,resultingingreaterpredictionaccuracyacrosstools.Furthermore,theR^2 scores for the FNN model improved for all seven tools, reflecting enhanced model fit and a better abilitytoexplainvarianceinthetargetvariable.ThelargeincreasesinR^2,particularlyfortools with low single-tool training performance, suggest that the dual training approach helps the FNN model capture important patterns that are not present when training on single-tool data alone.This improvement can be attributed to FNN's sensitivity to additional data,as the model's continuous learningstructureallowsittobettergeneralizeandreduceoverfittingwhenexposedtodiverse datasets.

The XGBoost model exhibited a similar trend, with MAE reductions observed for five tools andincreasesfortwotools,withameanchangeof-1.20,whichisasmalleraveragedecrease thanobservedintheFNNmodel.Thissmallerdecreasemaybeduetoseveralfactors.Firstly, XGBoost'ssingle-tooltrainingperformancewasalreadyhigh,leavinglessroomfornoticeable improvement through data aggregation.Additionally, XGBoost's tree-based architecture, with its strongregularizationmechanismsandrobustsplittingcriteria,inherentlyminimizesoverfitting, makingitlesssensitivetothebenefitsofadditionalsyntheticoraggregateddata.Ontheother hand, the average increase in MAE for the two tools was 0.92, lower than that observed in the FNN model, suggesting that while the gains from data augmentation were smaller, XGBoost maintained more stable and consistent performance across different tools.

IntermsofR^2scores,XGBoostshowedstrongoverallimprovements,withincreasesinsix out of seven tools.The only tool with a decrease experienced a minor drop of -0.03, indicating that

dual training generally enhances the model's ability to capture variance, although not as uniformly improving as observed in the FNN model.The improvements in R^2suggest that data aggregation still contributes to better generalization for XGBoost, even though its architecture limits the extent of performance changes compared to FNN.

These findings highlight the importance of data aggregation in improving model performance in regression tasks, particularly when single-tool training data is insufficient to capture underlying patterns.Whilebothmodelsbenefitedfromdualtraining,FNNexhibitedgreaterperformance gains, particularly in reducing MAE and enhancing R^2scores.This difference can be attributed to themodels'architectures:FNNmodelsrelyheavilyoncomplexfeatureinteractionsandbenefit from additional data to reduce overfitting and improve generalization.In contrast, XGBoost's tree-based ensemble structure, with its inherent robustness to data variability and strong regularization, makesitlesssensitivetothesizeanddiversityofthetrainingdata.Overall,FNNrequiresmore extensivedata(throughdataaggregation)tofullyextractmeaningfulpatterns,whileXGBoost maintainsstableperformanceevenwithlimiteddata,explainingthemoremodestbenefitsfrom data aggregation.

3.2.2 Linear Mixup Data Augmentation

The results for the slider datasets augmented by the two Mixup schemes previously discussed, one with 100% augmentation and one with 200% augmentation,are presented in Figure 3.6 and Fig-ure 3.7.The plots primarily illustrate the impact of Mixup:bars are green if the change is positive (lower MAE, higher R^2) and red if the change is negative (higher MAE, lower R^2). The first scheme creates two artificial data points between two real data points, and the second scheme creates one data point between two real data points. Training remained as before with single-tool and dual-tool training.

Becausesliderproductiondatasetsclusterintothreemajorregions,thevirtualmetrology function needs to classify each data point correctly into their major regions and detect FAILS (act

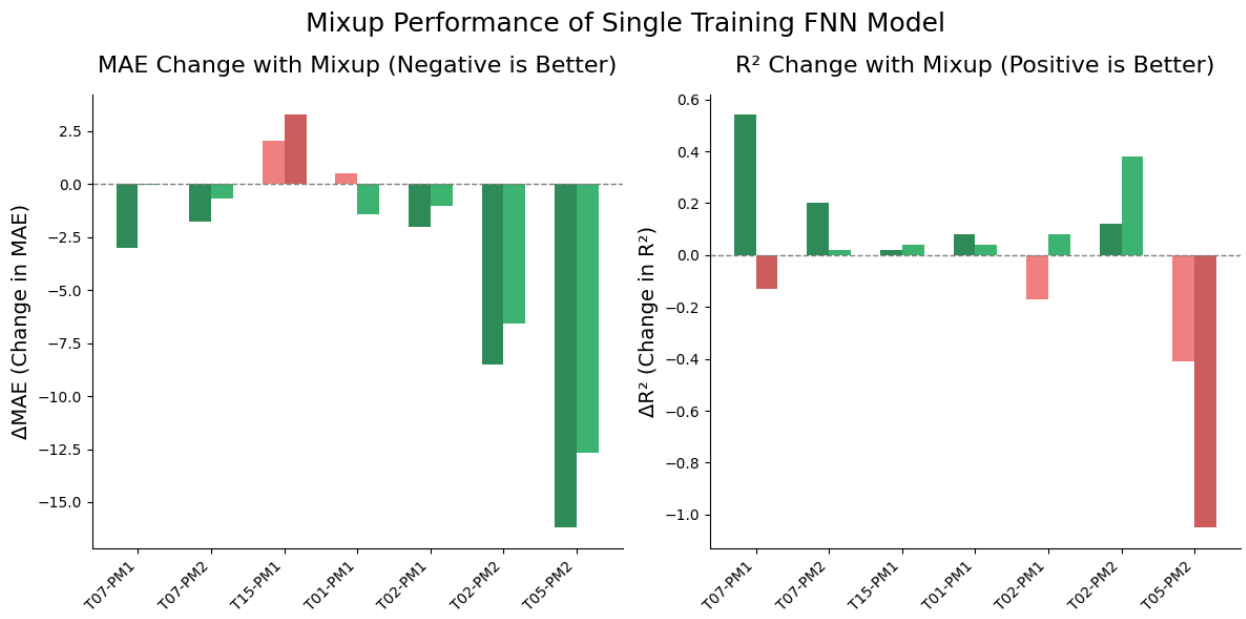

Figure3.6:MAEandR^2scoresfortheFNNmodeltrainedwithdatafromasingleslidertool usinglinearMixup.Ingeneral,thereisadecreaseinMAEandanincreaseinR^2scoresacross most slider tools.

likeoutlierpointsintrainingdata).Minimizingthemedianabsoluteerror(MAE)istherefore crucial to ensure predictions remain within the correct class region.Under the first Mixup scheme fortheFNNmodel,MAEdecreasedinfivetoolsandincreasedslightlyintwo,resultinginan averagechangeof-4.13(largestdecrease:16.2;largestincrease:2.06).Inthesecondscheme, MAE decreased for all but one tool (T01-PM1), with an average change of -2.73 (largest decrease: 12.7; largest increase: 3.27).The large decrease in MAE especially on T02-PM2 (close to -10) and T05-PM2 (around -15) can significantly improve the multi-class classification accuracy.Although R^2islessindicativeinexplainingperformanceofmulti-classclassification,itisstillusefulfor assessing the model's ability to capture overall variance.Specifically, the first scheme yielded an averageR^2increaseof0.05(largestincrease:0.54;largestdecrease:0.41),whereasthesecond scheme produced an average change of -0.09.The relatively small changes in R^2suggest that the original model may be adequately explain the variance in the target variable.

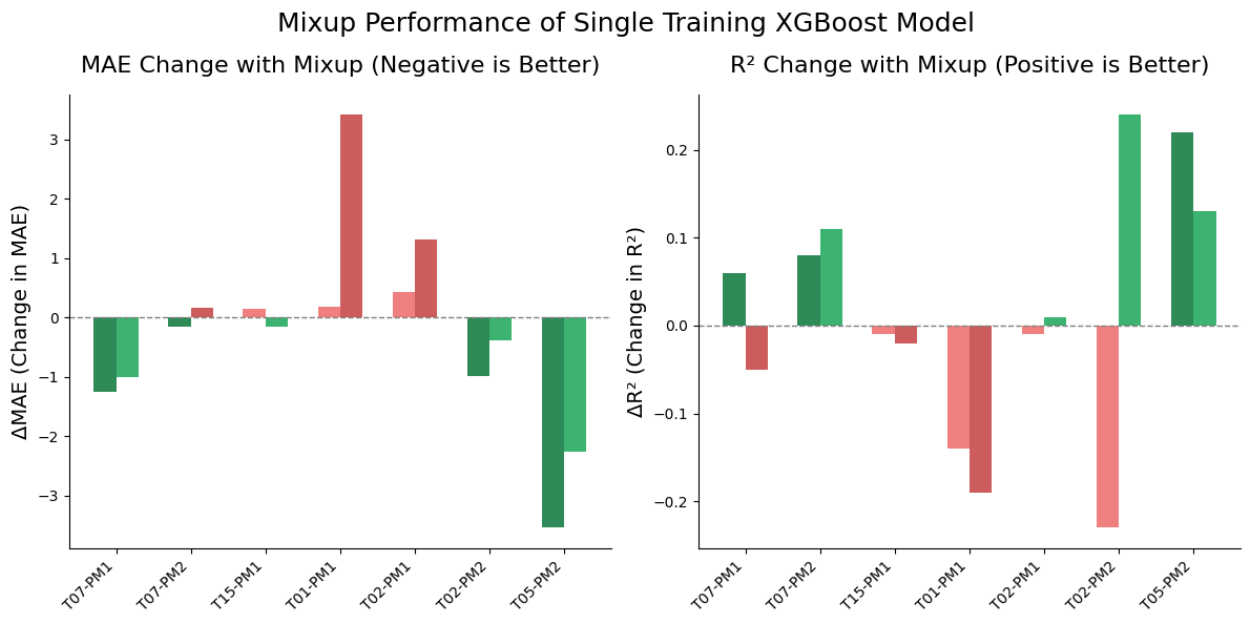

Figure 3.7:MAE and R^2scores for the XGBoost model trained with data from a single slider tool usinglinearMixup.SimilartrendsofdecreasedMAEandincreasedR^2scoresareobservedfor most tools, except for T05-PM2 where we saw a marked decrease in R^2scores.

Ontheotherhand,thegreatervariationinMAEsuggeststhatthemodel'sabsoluteerroris still influenced by the specific conditions of each tool.A higher variation in MAE could indicate that certain tools require additional data to improve prediction stability or that some regions in the featurespaceareunderrepresented.TheobservedreductionsinMAEacrossmosttoolsconfirm that the Mixup augmentation was generally beneficial,but the slight increases in MAE for a few tools highlight areas where additional adjustments, such as targeted data augmentation or feature refinement,maybeneeded.ForthefirstMixupschemewithXGBoost,MAEdecreasedinfour tools while increasing slightly in three, yielding an average change of -0.74 (largest decrease (posi-tive):3.54; largest increase (negative):0.42).The second Mixup scheme showed a similar pattern, withanaverageMAEchangeof0.15(largestdecrease(negative):2.27;largestincrease(posi-tive): 3.41, this scale of change in MAE won't have a significant impact on the model performance practically.IntermsofR^2,resultsweremixed:thefirstschemeimprovedscoresforthreetools

(averagechangeof-0.004),whilethesecondschemesawgainsinfourtools(averagechangeof 0.03).DespitetheseoutcomesXGBoostbenefitslessfromMixupcomparedtoFNN,primarily because Mixup generates linearly interpolated data that complements FNNs' continuous, gradient-based learning, helping smooth decision boundaries.By contrast, XGBoost relies on discrete tree splits,wheresuchinterpolationsofferlimitedvalueforsplitdecisions.Furthermore,XGBoost's strong built-in regularization reduces the added benefits of Mixup, while FNNs gain considerable regularization advantages, mitigating overfitting more effectively.

The results for both Mixup schemes on the FNN and XGBoost models using dual training are presented in Figure 3.8 and Figure 3.9.For both figures, each tool has two values for MAE and R^2, the darker hue being the first Mixup scheme and the second value being the second Mixup scheme. A green color indicates a positive change (like a decrease in MAE or increase in R^2) while a red color indicates a negative change (like an increase in MAE or decrease in R^2).

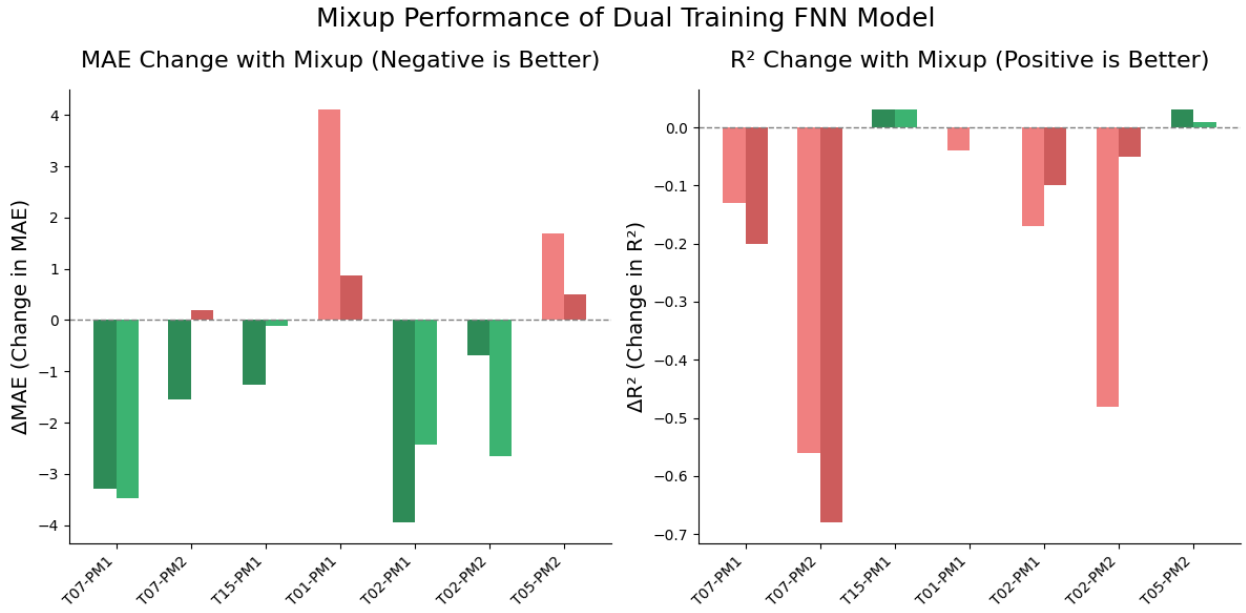

Figure 3.8:MAE and R^2scores for the FNN model trained with data from two slider tools using linear Mixup.There is a decrease in MAE in 5 tools and an increase in 2 tools for both schemes. R^2scores decreased in 5 tools for both schemes, with a slight increase in two tools.

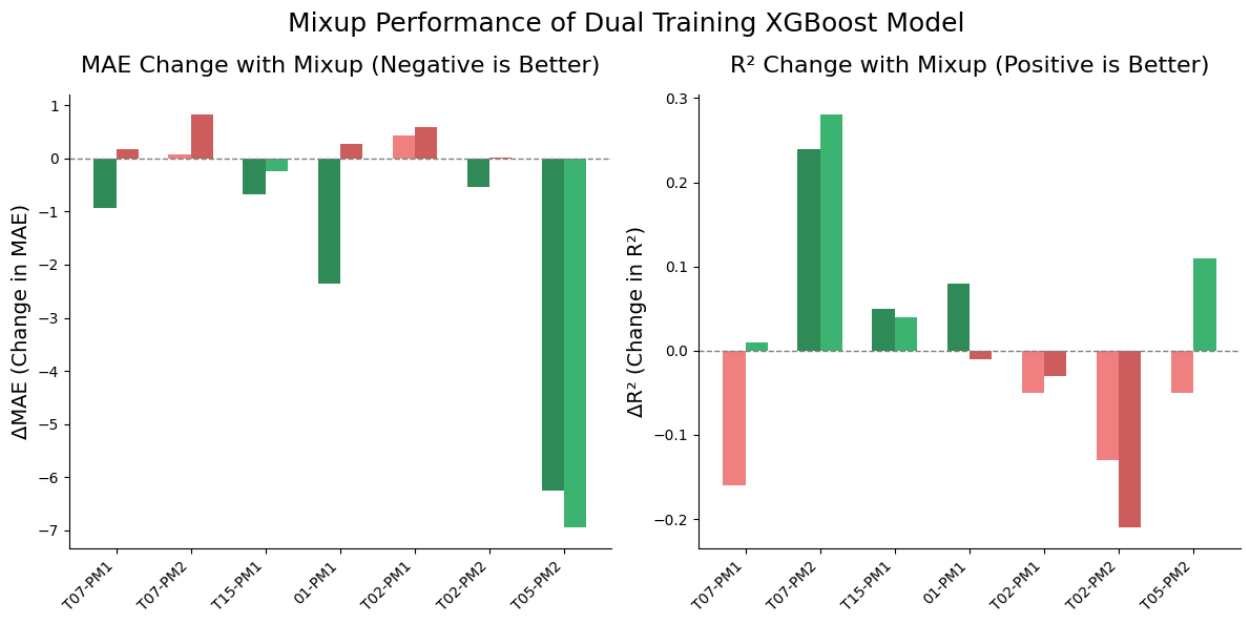

Figure3.9:MAEandR^2scoresfortheXGBoostmodeltrainedwithdatafromtwoslidertools using linear Mixup.MAE generally decreased across most tools, with increases being very slight. R^2scores saw a mixed pattern of increasing and decreasing scores.

InthefirstMixupschemefortheFNNmodel,MAEdecreasedforfivetoolsandincreased fortwo,averagingachangeof-0.71(largestdecrease:3.95;largestincrease:4.1).Underthe second scheme,MAE decreased in four tools and increased in three,yielding an average change of -1.01 (largest decrease:3.47; largest increase:0.87).Although the change in MAE is not large enough to significantly impact real-world model performance, it is still worth noting that Mixup in data aggregation has strong potential for improving model performance.Meanwhile, R^2declined infivetoolsandroseslightlyintwoforbothschemes,withaveragechangesof-0.19and-0.14, respectively,indicatingthatimprovementsinerrorreductiondidnotconsistentlytranslateinto higher overall variance capture.

UnderthefirstMixupschemewithXGBoost,MAEdecreasedinfivetoolswhilerising slightlyintwo,averagingachangeof-1.43(largestdrop:6.26;largestincrease:0.44).Inthe second scheme, MAE fell for two tools and rose for five, though some increases were under 1%.

Thelargestdecreasewas6.95,withanaveragechangeof–0.76.ThedecreaseofMAEonT05-PM2issignificantenoughtoimprovethemodelperformance,butthechangesinothertoolsare negligible.For R_2, the first scheme raised scores in three tools, resulting in an average change of-0.003, while the second scheme improved four tools with an average increase of 0.03.The largest R_2gain was 0.28, and the largest decrease was 0.21.

ThesummaryoftheoverallmodelperformancechangeresultsofMAEandR_2aboveis tabulated in table 3.1 and table 3.2.The first value in each cell is mean change, while the second value (blue) is median change:

Table 3.1:MAE Change

	FNN		XGB	
Scheme	Single	Dual	Single	Dua
Scheme 1	-4.13/-2	-0.71/-1.26	-0.74/-0.15	-1.43/-0.67
Scheme 2	-2.73/-1	-1.01/-0.12	0.15/-0.16	-0.76/0.18

Table 3.2:R2 Change

	FNN		XGB	
Scheme	Single	Dual	Single	Dual
Scheme 1	0.05/0.08	-0.19/-0.13	-0.004/-0.01	-0.003/-0.05
Scheme 2	-0.09/0.04	-0.14/-0.17	0.03/0.01	0.03/0.01

These results demonstrate the varying impact of linear Mixup data augmentation across differ-ent models and training schemes by demonstrating the mean and median value of change in MAE and R_2after Mixup.While improvements are observed in both metric scores in many cases, the ef-fectiveness of the Mixup technique appears to depend on both the model type and data aggregation status.

IntheFNNmodel,dualtrainingwithlinearMixupproducedmixedresults.Forthefirst scheme,MAEdecreasedacrossfivetoolswithanaveragereductionof17.0%,whileincreases

were observed in two tools, averaging 23.6%.Similarly, for the second scheme, MAE decreased byanaverageof16.6%infivetoolsandincreasedby4.4%intwotools.IntermsofR^2scores, the results were less favorable,with decreases in five tools and only slight increases in two tools under both schemes.The largest increase of R^2score in both schemes was 0.03 while the largest decrease was -0.68.These findings suggest that while Mixup can reduce errors, it may not always lead to improved model fit, particularly when the models are trained on data from two tools.

In contrast, the XGBoost model demonstrated more stable performance with Mixup applied under dual training.For the first scheme, MAE decreased across five tools by an average of 12.8%, while the increases, observed in two tools, averaged only 2.6%.The second scheme showed a less favorable balance, with MAE decreasing in two tools (average reduction of 11.4%) and increasing infivetools,thoughsomeincreaseswereminor(averageincreaseof3.8%).ForR^2,theresults showed mixed trends:under the first scheme,R^2increased for three tools and decreased for four tools,whereasinthesecondscheme,fourtoolssawanincreaseandthreetoolsexperienceda decrease.The largest increase of R^2score in both schemes was 0.28 while the largest decrease was-0.21.

The differences between the FNN and XGBoost models highlight the role of model architec-ture in how linear Mixup affects regression performance.FNN models appear to be more sensitive to both improvements and degradations in MAE and R^2, due to their reliance on complex feature relationships that can be influenced by synthetic data interpolation.XGBoost, with its tree-based architecture, shows smaller variations, due to its robust splitting criteria and regularization mecha-nisms.

According to Table 3.1 and Table 3.2, the impact of linear Mixup on regression performance differssignificantlybetweenFNNandXGBoostmodels.ForXGBoost,themeanandmedian differencesinMAEismuchsmallercomparedtoFNNforsingletraining,andslightlysmaller thanFNNfordualtraining;theR^2valuechangeisnegligible.ThesmallMAEandR^2changes indicate that Mixup has little effect on its performance.This can be attributed to XGBoost’s tree-

based architecture, which relies on robust splitting criteria and inherent regularization mechanisms that make it less sensitive to interpolated synthetic data.

In contrast, the FNN model shows more pronounced effects from Mixup.In single-tool train-ing,MixupleadstonotableimprovementsinbothmaeandR^2,withperformanceenhancingas moresyntheticdatapointsareadded(Scheme1).ThiscouldbeduetoMixupactingasanef-fective regularizer, improving generalization by encouraging the model to learn smoother decision boundariesandreducingoverfittingtonoisypatternsintheoriginaldata.However,indual-tool training,Mixupwithasmallersyntheticdataset(Scheme2)yieldsbetterresults.Thiscouldbe because Mixup operates directly at the data level, interpolating between samples from two poten-tially different data distributions.When the distributions diverge significantly, excessive synthetic datamayintroduceconflictingpatterns,makingitharderforthemodeltogeneralizeeffectively. Thus,asmallerMixupdatasetcanpreservethedistinctcharacteristicsofeachtool’sdatawhile still providing regularization benefits to achieve lower MAE.

Chapter 4

Conclusion

Thispaperpresentsananalysisofthedataqualityandengineeringrequirementsformachine learning-basedvirtualmetrologyforcommonmachinetoolsinthesemiconductorindustry.The virtual metrology application studied was the binary PASS/FAIL classification of the product qual-ity at the end of a machine tool step.Two multi-line, multi-machine semiconductor manufacturing operationswerestudied:fiveplasmaetchingtoolsusedinseparatewaferproductionlinesand sevenmillingtoolsusedinseparatesliderproductionlines.Thestudyfocusedonoptimumpro-cessing of the data by considering the conditions, functions, and data requirements of the machines together and how to leverage commonality.

FeedforwardNeuralNetwork(FNN)andXGBoostalgorithmswereusedandcomparedfor boththewaferandsliderproductionmachines.Forwaferproductionthealgorithmswereap-plied directly for PASS/FAIL classification.For slider production the algorithms were formulated asregressionmodelstopredictproductthicknessthatstratifiedintothreeclustersfordifferent products.Outlier identification was used to then classify PASS or FAIL. The following data pro-cessing/engineering approaches were examined:

- Data aggregation across one or more machines to increase variation and operational cover-age.

•Dimensionality reduction techniques were applied to reduce noise and enhance feature extraction.

•Aseparatescalingapproachwasimplementedduringdataaggregationtoimprovemodel performancebyaligningthefeaturedistributionsofdifferentdatasetsandtomitigatediscrepanciescaused byvarying single-feature distributions despite similar feature dependencies.

•LinearMixupalgorithmwasappliedtoaugmentdataandaddressthescarcityofcollected data from slider production tools

•FNN with transfer learning with live model update was used for plasma etching to address process drift.Varying train/test and data length ratios were also tested.

Forthewaferproductionmachines,theXGBoostalgorithmdemonstratedbetterorcomparableperformancetoFNNinbothsingle-toolanddual-tooltrainingwhilerequiringfewercomputationalresourcesandofferinggreaterrobustness,makingitthebetterchoiceforthissolution objective.Transfer learning significantly improved model performance across all tested datasets, demonstratingtheeffectivenessofliveupdating.Notably,usingonly20%ofnewonlinedata forupdatesledtosubstantialperformancegains,reducingtheneedforfrequentofflinephysical measurements and associated costs.Transfer learning also overrode the advantages of data aggregation.Ingeneral,giventheoperationaldriftwiththewaferproductiontools,transferlearning proved to be a fruitful endeavor.

For the slider tools, XGBoost consistently outperformed or matched FNN in regression tasks in terms of Median Absolute Error (MAE) and R2.Given better performance for both slider and waferproduction,XGBoostdemonstratedversatilityforbothclassificationandregressiontasks. Due to its tree-based structure and inherent regularization from ensemble learning, XGBoost was less sensitive to data aggregation compared to the FNNs, though data aggregation still yielded pos-itive effects across all datasets.Mixup strategies demonstrated notable performance improvements

for the FNN models, particularly for single-tool training, due to its regularization effect which pro-moted better generalization. In contrast, Mixup impact on XGBoost was minimal likely because of XGBoost's tree-based architecture and robust splitting criteria.With respect to dual-tool aggrega-tion, FNN performed better with smaller synthetic datasets, possibly due to distribution differences between datasets that can introduce conflicting patterns when overly mixed.For XGBoost, modest MAEimprovementswereobserved,indicatingaresiliencetodatavariations.Overall,theeffec-tiveness of Mixup depends on model architecture and data characteristics.

While no single method universally outperforms others, the combination of Mixup, data ag-gregation, and transfer learning generally enhanced model performance.When systematically ap-plied as a data engineering procedure, the combination of methods offered significant improvement to this in industrial virtual metrology PASS/FAIL application.

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